We find that truncating the update equations to $\mathcal{O}(\varepsilon^3)$ is sufficient to explicitly resolve the leading order behavior of the error for these discretization schemes, and we give the differences between observed numerical averages and exact averages below:

$$\begin{aligned}
\langle q^2\rangle_h^{[\![\text{ABOBA}]\!]} - \langle q^2\rangle &= \frac{3h^2\varepsilon\left[48\varepsilon - 4 + 3\gamma^2(9\varepsilon - 1)\right]}{12\gamma^2 + 16} + \mathcal{O}(\varepsilon h^4 + \varepsilon^3 h^2),\\
\langle q^2\rangle_h^{[\![\text{BAOAB}]\!]} - \langle q^2\rangle &= -\frac{9h^2\varepsilon^2}{6\gamma^2 + 8} - \frac{3h^4\varepsilon^2\left[6\gamma^4 + 19\gamma^2 + 16\right]}{8\left(3\gamma^2 + 4\right)^2} + \mathcal{O}(\varepsilon^2 h^6 + \varepsilon^3 h^2),\\
\langle q^2\rangle_h^{[\![\text{OBABO}]\!]} - \langle q^2\rangle &= h^2\left(\frac{1}{4} + \frac{\varepsilon\left[144\varepsilon + 81\varepsilon\gamma^2 - 9\gamma^2 - 12\right]}{12\gamma^2 + 16}\right) + \mathcal{O}(h^4 + \varepsilon^3 h^2),\\
\langle q^2\rangle_h^{[\![\text{OABAO}]\!]} - \langle q^2\rangle &= -h^2\left(\frac{1}{4} + \frac{18\varepsilon^2}{12\gamma^2 + 16}\right) + \mathcal{O}(h^4 + \varepsilon^3 h^2).
\end{aligned} \tag{7.14}$$

The behavior of the methods can thus be summarized as

$$\begin{aligned}
\left|\langle q^2\rangle_h^{[\![\text{ABOBA}]\!]} - \langle q^2\rangle\right| &= O\left(\varepsilon h^2\right),\\
\left|\langle q^2\rangle_h^{[\![\text{BAOAB}]\!]} - \langle q^2\rangle\right| &= O\left(\frac{\varepsilon^2 h^2}{1+\gamma^2} + \varepsilon^2 h^4\right),\\
\left|\langle q^2\rangle_h^{[\![\text{OBABO}]\!]} - \langle q^2\rangle\right| &= O\left(h^2\right),\\
\left|\langle q^2\rangle_h^{[\![\text{OABAO}]\!]} - \langle q^2\rangle\right| &= O\left(h^2\right).
\end{aligned} \tag{7.15}$$

We would expect that both the $[\![\text{ABOBA}]\!]$ and $[\![\text{BAOAB}]\!]$ schemes would have a leading error term proportional to ε, given that it was shown in Table 7.1 that in the pure harmonic case (at $\varepsilon = 0$) these methods give perfect sampling. The twofold surprising result though, is that (a) the leading error of the $[\![\text{BAOAB}]\!]$ scheme has behavior proportional to ε^2, while (b) the leading order term of the $[\![\text{BAOAB}]\!]$ scheme is annihilated for large γ.

It would seem that by ordering the A, B and O update terms favourably in an algorithm we are able to achieve significant sampling improvements on systems that are "nearly harmonic", at no extra cost (in terms of force evaluations).

To illustrate this numerically, we use each of the four given second order methods to compute the absolute error in $\langle q^2\rangle_h$ using the one dimensional perturbed harmonic oscillator with Hamiltonian given in (7.13) at a range of timesteps $h \in [0.1, 2]$ and nonlinearity parameters $\varepsilon \in [0.0001, 1]$. We choose $\beta = \gamma = 1$ for all experiments, and run the schemes using each parameter set for a fixed time window of length 10^8. The results are shown in Fig. 7.3, and qualitatively agree with the results of (7.15).